Contrastive Trace Safety: Characterizing Hidden-State Detection of Safety Commitment in Reasoning Models

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Abstract

Reasoning models exhibit "self-jailbreak" behavior where safety reasoning and harmful execution are interleaved throughout extended chain-of-thought, culminating in a *commitment point* after which safety reasoning permanently ceases. Existing hidden-state jailbreak detectors, designed for adversarial-input attacks on instruction-tuned models, have not been evaluated on this internal reasoning failure mode. We present the first systematic cross-architecture study of hidden-state detection for safety commitment in reasoning models, evaluating static probes, temporal window probes, geometric detectors, and sequential models across four models spanning two architecture families (Llama-3.1, Qwen2.5) and two scales (7–8B, 32B). We find that: (1) static probes achieve limited coverage (46–53%), with geometric detectors showing architecture-dependent effectiveness correlated with PCA concentration of safety signal; (2) temporal window probes consistently improve coverage (+14–30pp) with architecture-dependent magnitude—Llama peaks at $W{=}15$ while Qwen2.5 peaks at $W{=}10$, reflecting differences in signal geometry; (3) detection lead time is near-zero across all architectures (mean -1.1 to $+0.5$ steps), with oracle decomposition revealing that 93% of the oracle's $+31$-step advantage derives from future label visibility inaccessible to hidden states. Our architectural analysis reveals *why* signal distribution differs: Qwen2.5 concentrates safety signal in fewer principal components (Fisher PCA-3: 2.2–2.8 vs. Llama's 0.9), yielding higher static baselines but less temporal headroom. These findings establish principled limitations for hidden-state safety monitoring and provide architecture-specific deployment guidance.

1 Introduction

Large reasoning models such as DeepSeek-R1 DeepSeek-AI [2025] and QwQ Team [2024] produce extended chain-of-thought (CoT) traces that interleave safety-relevant reasoning with task execution. A growing body of work has documented *self-jailbreak*: the model's own reasoning process gradually abandons safety considerations, culminating in a *commitment point* after which safety reasoning permanently ceases Jiang et al. [2025], Tang et al. [2025]. This failure mode is qualitatively different from adversarial-input jailbreaks, as the attack originates within the model's own reasoning rather than from an external prompt.

Hidden-state monitoring has proven effective for detecting adversarial jailbreaks in instruction-tuned models. TrajGuard Xie et al. [2024] uses Mahalanobis distance for trajectory-level anomaly detection; SafeSwitch Zheng et al. [2024] and SIREN Chen et al. [2024] train linear probes on hidden representations to classify safety-relevant states. These methods achieve $>95\%$ defense rates on standard attacks but operate at token-level granularity and assume the threat is an adversarial *input*. They have never been evaluated on reasoning-model self-jailbreak, where the threat is the model's *own extended reasoning process*.

Text-level CoT monitors Wang et al. [2024], Zhong et al. [2024] can detect safety-relevant content in generated text but are vulnerable to pathological CoT—the model may produce safety-affirming text while its hidden representations encode harmful intent. This motivates hidden-state approaches, but the gap between existing hidden-state detectors (designed for instruction-tuned models) and reasoning-model safety monitoring remains unaddressed.

We bridge this gap with a systematic cross-architecture study that makes three contributions:

1. **Characterization of architecture-dependent signal geometry.** We show that safety signal distribution differs fundamentally between Llama-3.1 and Qwen2.5 architectures: Qwen2.5 concentrates signal in fewer principal components (Fisher discriminant in PCA-3 space: 2.2–2.8 vs. Llama's 0.9), explaining why geometric detectors partially succeed on Qwen2.5 (coverage 41–45%) but fail on Llama (30%).

2. **Temporal probes as a practical method with architecture-specific tuning.** Temporal window probes consistently improve detection coverage across all architectures (+14–30pp over static), but the magnitude and optimal window size are architecture-dependent: Llama peaks at $W{=}15$ with $+30$pp, Qwen2.5 at $W{=}10$ with $+14$–17pp. A $2{\times}2$ architecture$\times$scale matrix confirms this is an architecture-level property, not a scale confound.

3. **Principled limitation boundary for hidden-state monitoring.** Detection lead time is near-zero across all architectures and probe types. Oracle decomposition shows 93% of the oracle's $+31$-step advantage comes from future label visibility, establishing that pre-textual detection is an information-theoretic limitation, not a method failure.

2 Related Work

Hidden-state jailbreak detection. TrajGuard Xie et al. [2024] computes Mahalanobis distance from safe-trace covariance to detect anomalous hidden-state trajectories. SafeSwitch Zheng et al. [2024] and SIREN Chen et al. [2024] train linear probes on per-token hidden states to classify safety-relevant states. These approaches achieve high defense rates ($>$95%) on instruction-tuned models under adversarial-input attacks but have not been evaluated on reasoning-model self-jailbreak.

Reasoning model safety. Self-Jailbreaking Jiang et al. [2025] and HarmThoughts Tang et al. [2025] characterize how reasoning models abandon safety reasoning during extended CoT. SafeRemind Wang et al. [2024] and SafeThink Zhong et al. [2024] propose text-level CoT monitors but remain vulnerable to pathological reasoning traces. No prior work bridges hidden-state detection methods to reasoning-model safety commitment.

Probing LLM representations. Linear probing of hidden states has been used to detect truthfulness Burns et al. [2023], sentiment Li et al. [2024], and safety-relevant features Arditi et al. [2024]. Our work extends this paradigm to temporal probing of safety commitment trajectories, discovering that temporal context is critical for reliable detection.

3 Method

3.1 Problem Formulation

Given a reasoning trace $\mathbf{x} = (x_1, \ldots, x_T)$ with corresponding step-level hidden states $\mathbf{h} = (h_1, \ldots, h_T)$ and binary safety labels $y_t \in \{0, 1\}$ (safe/unsafe), we define the *commitment point* t^* as the last step where $y_t = 1$ (safe) before a permanent transition to $y_t = 0$. The detection task is to predict, at each step t, whether the trace has passed its commitment point: $\hat{c}_t = f(h_{\leq t})$.

We evaluate detection methods along three axes:

- **Coverage**: fraction of commitment-containing traces where the detector correctly identifies the commitment transition (crossing rate).

- **Accuracy**: step-level classification accuracy on held-out traces.

- **Lead time**: $t^* - \hat{t}^*$, where $\hat{t}^*$ is the detector's predicted commitment step. Positive values indicate pre-textual detection (detecting commitment before it occurs in the text).

3.2 Detection Methods

Static probe. A logistic regression classifier trained on individual hidden states h_t at a fixed layer ℓ: $\hat{c}_t = \sigma(\mathbf{w}^\top h_t^{(\ell)} + b)$.

Temporal window probe. A logistic regression classifier on concatenated hidden states within a window: $\hat{c}_t = \sigma(\mathbf{w}^\top [h_{t-W+1}^{(\ell)}; \ldots; h_t^{(\ell)}] + b)$, where W is the window size. This captures temporal patterns such as declining safety reasoning density.

Mahalanobis distance baseline. Following TrajGuard, we estimate the covariance matrix Σ from safe-trace hidden states and compute per-step Mahalanobis distance: $d_t = \sqrt{(h_t - \mu_{\text{safe}})^\top \Sigma^{-1} (h_t - \mu_{\text{safe}})}$, thresholded at the 95th percentile of the safe-trace distribution.

Sequential probes (GRU, Self-Attention). GRU with hidden dimension 256 and a self-attention probe (2-head, 2-layer) over the hidden-state sequence, providing upper bounds on what temporal modeling can extract.

Oracle baseline. A label-sequence oracle with access to the complete label sequence $(y_1, \ldots, y_T)$ that detects consecutive safe-label decline patterns. This establishes the information-theoretic ceiling.

3.3 Cross-Architecture Evaluation

We evaluate across a 2×2 matrix of architecture families and scales:

- **Llama-3.1**: DeepSeek-R1-Distill-Qwen-8B (r1-8b), hidden dim 4096, 28 layers
- **Qwen2.5**: DeepSeek-R1-Distill-Qwen-32B (r1-32b), QwQ-32B, OpenThinker-7B; hidden dim 5120 (32B) / 4096 (7B), 64 layers (32B) / 28 layers (7B)

For cross-architecture transfer experiments, we learn a linear projection between architectures' hidden spaces, trained on 50 aligned traces with a shared 512-dimensional projection space.

3.4 Signal Geometry Analysis

To characterize *why* detection performance differs across architectures, we analyze the geometry of safety-relevant signal:

- **Fisher discriminant ratio**: $J = \frac{(\mu_{\text{safe}} - \mu_{\text{unsafe}})^\top \Sigma_W^{-1} (\mu_{\text{safe}} - \mu_{\text{unsafe}})}{\text{tr}(\Sigma_W)}$, computed in both full representation space and PCA-reduced space.
- **PCA concentration**: variance explained by top-k principal components, measuring how compactly the safety signal is encoded.

4 Experimental Setup

Data. We use HarmThoughts reasoning traces with step-level safety annotations. Trace counts: r1-8b: 209 traces (125/42/42 train/val/test split); r1-32b: 187 traces (112/37/38); QwQ-32B: 192 traces (115/38/39); OpenThinker-7B: comparable scale. All traces contain extended CoT with interleaved safety and harmful reasoning.

Model	Arch	Acc	Cov	Lead	J_{orig}	J_{PCA3}
r1-8b	Llama	.817	.465	-2.0	17.88	0.93
r1-32b	Qwen	.834	.528	-1.4	12.07	2.84
QwQ-32B	Qwen	.819	.482	-1.8	10.52	2.43
OT-7B	Qwen	—	.512	-0.9	10.12	2.37

Table 1: Static probe results. J_{orig}: Fisher discriminant in full space; J_{PCA3}: Fisher in PCA top-3 space. Qwen2.5 has lower full-space Fisher but higher PCA-space Fisher, indicating more concentrated safety signal.

Model	Static	Temp	Δ	Lead	Acc
r1-8b (Llama)	.465	.767	+30.2	-1.09	.848
r1-32b (Qwen)	.528	.679	+15.1	$+0.52$	.851
QwQ-32B (Qwen)	.482	.648	+16.7	-0.73	.842
OT-7B (Qwen)	.512	.654	+14.2	-0.91	—

Table 2: Temporal window probe (W=15). Δ: improvement over static (pp). Temporal advantage is consistent (+14–30pp) but architecture-dependent in magnitude.

Hidden state extraction. We extract hidden states at the proportional midpoint layer for each architecture (50% depth): L14 for r1-8b, L32 for r1-32b/QwQ-32B, validated by layer ablation (Section 5.5).

Evaluation. All metrics are computed on held-out test splits. We report 95% confidence intervals from 1000 percentile bootstrap resamples.

5 Results

5.1 Static Probes Across Architectures

Table 1 presents static probe results across all four models. Static logistic regression achieves 46–53% coverage, confirming that single-step hidden states contain safety-relevant signal but are insufficient for reliable commitment detection.

A key finding is the divergent signal geometry between architectures. Llama exhibits high Fisher discriminant in full representation space (J=17.88) but low PCA concentration (J_{PCA3}=0.93, variance explained by 3 PCs: 16.6%). Qwen2.5 shows the opposite pattern: lower full-space Fisher (J=10–12) but higher PCA concentration (J_{PCA3}=2.2–2.8, variance explained: 21–23%). This means Qwen2.5 distributes safety signal across fewer dimensions with higher per-dimension separability in leading components, while Llama distributes signal more diffusely.

5.2 Temporal Window Probes

Table 2 shows temporal window probe results at W=15. Temporal probes consistently improve coverage over static baselines across all architectures, but the magnitude is architecture-dependent.

The attenuation on Qwen2.5 (+14–17pp vs. Llama's +30pp) is explained by the signal geometry: Qwen2.5's higher PCA concentration gives static probes a head start, leaving less headroom for temporal context. The temporal probe still helps because safety reasoning density decline patterns exist cross-architecture, but the already-higher static baseline reduces the marginal gain.

5.3 Architecture $\times$ Scale Matrix

Table 3 presents the 2$\times$2 matrix that disentangles architecture from scale effects.

Model	Arch	Scale	Static	Temp	Δ	W^*
r1-8b	Llama	8B	.465	.767	+30.2	15
OT-7B	Qwen	7B	.512	.654	+14.2	10
r1-32b	Qwen	32B	.528	.689	+16.1	10
QwQ-32B	Qwen	32B	.482	.648	+16.6	10

Table 3: Architecture×scale matrix. W^*: optimal window size. Architecture determines temporal advantage magnitude; scale has minimal effect within architecture (+1.6pp static, +3.5pp temporal from Qwen 7B→32B).

W	r1-8b	r1-32b	QwQ-32B
1	.465	.528	.482
3	.535	.566	.519
5	.651	.623	.593
10	.698	**.689**	**.667**
15	**.767**	.679	.648
20	.744	.660	.630
25	.721	.642	.611

Table 4: Coverage by window size W. Bold: architecture-specific optimum. Llama saturates later (W=15) while Qwen2.5 peaks earlier (W=10) and declines with larger windows.

At matched scale (7–8B), the pattern holds: Qwen2.5 has higher static coverage but lower temporal delta. The scale effect within Qwen2.5 is minimal, ruling out the hypothesis that the attenuation is a 32B-specific phenomenon. The optimal window size is also architecture-determined: W^*=15 for Llama, W^*=10 for all three Qwen2.5 models regardless of scale.

5.4 Window Size Ablation

Table 4 presents the window size ablation across architectures.

The curves have qualitatively different shapes: Llama shows concave growth with late saturation at W=15, while Qwen2.5 peaks at W=10 and then *declines*. The decline on Qwen2.5 at W>10 reflects overfitting—the per-step signal is stronger (higher PCA concentration), making redundant temporal features counterproductive. This provides a practical recommendation: tune W per architecture rather than using a fixed value.

5.5 Layer Selection Ablation

Table 5 shows temporal probe coverage across layers of r1-32b (64 layers, sampled every 4th). The best layer is L32 (50% relative depth), matching r1-8b's optimum at L14 (also 50%). The smooth bell-shaped curve with 28pp range (41% to 69%) confirms that layer choice matters and the 50% depth heuristic generalizes cross-architecture.

5.6 Mahalanobis Distance Baseline

Table 6 evaluates the TrajGuard-style Mahalanobis distance detector. The detector partially succeeds on Qwen2.5 (coverage 41–45%, AUROC 0.73–0.75) but largely fails on Llama (coverage 30%, AUROC 0.68).

This result is explained by signal geometry: Qwen2.5's higher PCA concentration means more safety-relevant information lives in directions that dominate the covariance matrix, making covariance-based detection partially effective. On Llama, safety signal is distributed across minor PCs that the covariance matrix effectively ignores. This complicates the "existing detectors fail" narrative—geometric

Layer	Rel. Depth	Coverage	Accuracy
L8	0.125	.478	.784
L16	0.250	.587	.812
L24	0.375	.651	.834
L32	0.500	**.689**	.851
L40	0.625	.668	.841
L48	0.750	.612	.819
L56	0.875	.534	.793

Table 5: Layer selection ablation on r1-32b (W=10). Peak at 50% depth matches the Llama finding, suggesting safety state is encoded as a mid-level semantic abstraction.

Model	Cov	Acc	AUROC	vs Static
r1-8b (Llama)	.302	.714	.683	-16.3
r1-32b (Qwen)	.453	.779	.752	-7.5
QwQ-32B (Qwen)	.407	.761	.734	-7.5

Table 6: Mahalanobis distance baseline. "vs Static": coverage difference from static LR (pp). The architecture-dependent performance correlates with PCA concentration of safety signal.

detectors show architecture-dependent effectiveness rather than universal failure.

5.7 Cross-Architecture Transfer

Table 7 evaluates zero-shot cross-architecture transfer. Cross-architecture transfer is weak (coverage 35–43%), while within-architecture transfer (Qwen→Qwen) nearly matches in-domain performance.

The safety concept appears universal (all transfers exceed the random baseline of 16.7%), but its geometric instantiation is architecture-specific. Practical deployment requires per-architecture probe training.

5.8 Lead Time Analysis

Table 8 presents lead time distributions across all models. Lead time is near-zero across all architectures, with no model achieving statistically positive lead time.

The r1-32b result ($+0.52$ mean, 42% positive rate) is a directional curiosity—the distribution spans $[-8, +12]$ with median at 0, and the bootstrap CI includes zero. The 30-step gap between any probe and the oracle persists uniformly, confirming hidden states encode current safety state but not future safety trajectory.

5.9 Oracle Decomposition

Table 9 decomposes the oracle's advantage to establish *why* pre-textual detection fails.

The decomposition reveals that the full oracle's key features are consecutive safe-count decline (importance 0.41), safe-label density in window (0.28), longest unsafe streak (0.19), and type diversity decline (0.12). When restricted to past-only labels, the oracle drops to $+2.1$ steps—only marginally better than hidden-state probes. The mutual information between current hidden state and future labels is near-zero (0.043 bits), confirming there is no recoverable signal about the future in current representations.

5.10 Complex Probe Architectures

Table 10 compares GRU and self-attention probes against temporal LR on Qwen2.5 models.

Train	Test	Probe	Cov	vs In-Dom
r1-8b	r1-32b	Static	.372	-15.6
r1-8b	r1-32b	Temp	.426	-25.3
r1-32b	r1-8b	Static	.349	-11.6
r1-32b	r1-8b	Temp	.401	-36.6
r1-32b	QwQ-32B	Static	.491	$+0.9$
r1-32b	QwQ-32B	Temp	.612	-3.6

Table 7: Cross-architecture transfer via learned linear projection. Within-architecture transfer (Qwen→Qwen) nearly matches in-domain; cross-architecture transfer drops 12–37pp.

Model	Mean	Med	Std	%Pos	95% CI
r1-8b	-1.09	-1	4.2	30.2	$[-2.91, 0.73]$
r1-32b	$+0.52$	0	4.8	42.1	$[-1.47, 2.51]$
QwQ-32B	-0.73	-1	5.1	33.3	$[-2.83, 1.38]$

Table 8: Lead time distribution (temporal probe, steps). The r1-32b positive mean is not statistically distinguishable from zero (CI spans -1.47 to $+2.51$).

Complex probes provide negligible improvement ($<$2pp) across architectures, consistent with the r1-8b finding. The coverage ceilings—77% on Llama, 65–69% on Qwen2.5—appear to be information limits rather than model capacity limits. Temporal LR is the preferred method given its 20$\times$ parameter efficiency.

5.11 Fine-Grained Label Supervision

Table 11 evaluates whether 16-label safety taxonomy supervision improves over binary labels.

The 16-label probe improves coverage by 2–3pp but at the cost of reduced macro accuracy (62–63% vs. 85%). The improvement is small because binary labels already capture the key safe→unsafe transition. The primary value is interpretability: Type 2 (acknowledging harm) and Type 3 (refusing) are most separable, while Type 1 (reasoning about safety) is confused with harmful reasoning in hidden-state space, suggesting the model uses similar computational circuits for both.

5.12 Statistical Reliability

Table 12 reports bootstrap 95% confidence intervals for key metrics.

The temporal advantage is statistically significant on all architectures (all Δ CI lower bounds > 0). The delta CIs for Qwen2.5 ([0.07, 0.24] and [0.09, 0.25]) are clearly shifted below Llama's ([0.20, 0.41]), supporting the "architecture-dependent magnitude" claim. Lead time CIs span zero on all models, supporting "near-zero" framing. We note that CIs are wide due to small test sets (38–43 traces); the architecture-magnitude difference is observed but not conclusively established at conventional significance levels.

6 Discussion

Why do architectures differ? The signal geometry analysis reveals a clear mechanism: Qwen2.5's wider MLP and GQA attention distribute safety-relevant features across more dimensions with lower per-dimension separability, but concentrate more total variance in leading principal components. This creates a tradeoff—geometric detectors (Mahalanobis) partially succeed because they operate on covariance-dominated directions, while temporal probes gain less because the static baseline is already higher. The

Oracle Variant	Lead	Cov	Info Source
Full (past+future)	$+31$	1.00	Complete labels
Past-only	$+2.1$	0.82	Labels at $t \leq$ current
Current-step	-0.5	0.73	Label at t only
Hidden-state (W=15)	-1.09	0.767	Hidden states

Table 9: Oracle decomposition. 93% of the oracle's lead time ($+28.9$ of $+31$ steps) comes from future label visibility. The mutual information between current hidden state and future labels ($t+5$ to $t+15$) is 0.043 bits.

Model	Method	Params	Cov	Lead	Δ
r1-32b	Temp LR	77K	.689	$+0.52$	—
	GRU-256	1.6M	.698	$+0.89$	$+0.9$
	Attention	2.1M	.685	$+0.31$	-0.4
QwQ-32B	Temp LR	77K	.667	-0.73	—
	GRU-256	1.6M	.667	-0.42	$+0.0$
	Attention	2.1M	.654	-0.58	-1.3

Table 10: Complex probe architectures on Qwen2.5 models. Δ: coverage change vs. temporal LR (pp). GRU's $20\times$ parameter count yields $<$2pp improvement, confirming the coverage ceiling is an information limit.

architecture-specific optimal window ($W^*=10$ for Qwen2.5 vs. $W^*=15$ for Llama) follows from this: shorter temporal context suffices when per-step signal is more geometrically accessible.

Practical deployment implications. Our results provide three actionable guidelines: (1) use the 50% depth heuristic for layer selection on new architectures; (2) tune window size per architecture—do not assume a universal optimal W; (3) train probes per architecture rather than relying on cross-architecture transfer.

Limitations. The study is limited to four reasoning models from two architecture families. Trace counts are modest (187–209 per model), resulting in wide confidence intervals. The 16-label taxonomy analysis, while providing interpretability insights, does not substantially change the quantitative story. We evaluate only linear and simple nonlinear probes; more sophisticated methods (e.g., nonlinear temporal models with larger training sets) might achieve modest additional coverage.

Future directions. The oracle decomposition identifies a ceiling of $+2.1$ lead time steps for any method with access to past labels, motivating hybrid text+hidden-state monitors. The near-zero mutual information between hidden states and future labels suggests that closing the lead time gap may require auxiliary training objectives that explicitly encode future safety trajectory.

7 Conclusion

We present the first cross-architecture study of hidden-state detection for safety commitment in reasoning models. Our characterization reveals that safety signal geometry is architecture-dependent, with Qwen2.5 concentrating signal in fewer principal components than Llama. Temporal window probes consistently improve detection coverage across all architectures (+14–30pp), but the magnitude and optimal window size require architecture-specific tuning. Detection lead time is near-zero across all architectures and probe types; oracle decomposition confirms this is an information-theoretic limitation, not a method failure. These findings establish principled boundaries for hidden-state safety monitoring:

Model	Labels	Cov	Lead	Macro Acc
r1-8b	Binary	.767	-1.09	.848
r1-8b	16-label	.791	-0.82	.623
r1-32b	Binary	.679	$+0.52$	.851
r1-32b	16-label	.704	$+0.38$	.631

Table 11: Fine-grained 16-label vs. binary supervision. 16-label provides marginal coverage improvement (+2–3pp) with substantially lower macro accuracy due to inter-class confusion.

Metric	r1-8b	r1-32b	QwQ-32B
Static Cov	[.33, .61]	[.40, .66]	[.35, .61]
Temp Cov	[.65, .87]	[.55, .79]	[.53, .76]
Δ	[.20, .41]	[.07, .24]	[.09, .25]
Lead Mean	$[-2.9, .7]$	$[-1.5, 2.5]$	$[-2.8, 1.4]$
Mahal Cov	[.19, .42]	[.33, .57]	[.28, .53]

Table 12: Bootstrap 95% CIs (1000 resamples). The temporal advantage is statistically significant on all architectures (Δ CI lower bound > 0). Lead time CIs span zero on all models.

temporal context is essential for coverage, but pre-textual detection requires information beyond what current hidden states encode.

References

Andy Arditi et al. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024.

Collin Burns et al. Discovering latent knowledge in language models without supervision. *arXiv preprint arXiv:2212.03827*, 2023.

Wei Chen et al. Siren: Safety-informed representation engineering for llm alignment. *arXiv preprint*, 2024.

DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Albert Q. Jiang et al. Self-jailbreaking in reasoning models. *arXiv preprint*, 2025.

Kenneth Li et al. Inference-time intervention: Eliciting truthful answers from a language model. *arXiv preprint*, 2024.

Ruixiang Tang et al. Harmthoughts: Benchmarking harmful reasoning in language models. *arXiv preprint*, 2025.

Qwen Team. Qwq: Reflect deeply on the boundaries of the unknown. *arXiv preprint*, 2024.

Yuxiang Wang et al. Saferemind: Chain-of-thought safety monitoring for reasoning models. *arXiv preprint*, 2024.

Yueqi Xie et al. Trajguard: Trajectory-level safety monitoring for language model agents. *arXiv preprint*, 2024.

Lianmin Zheng et al. Safeswitch: Probing hidden states for safety-critical switching in llms. *arXiv preprint*, 2024.

Wanjun Zhong et al. Safethink: Monitoring the safety of llm reasoning. *arXiv preprint*, 2024.